Umesh Kashyap

umeshk@iitbhilai.ac.in Bhilai, India LinkedIn GitHub Google Scholar

PROFILE SUMMARY

I'm a Ph.D. Scholar in the **Department of Computer Science and Engineering** at the **Indian Institute of Technology (IIT) Bhilai**. My research focuses on **Trustworthy AI, Adversarial Machine Learning, and AI Security**, particularly on developing robust defenses against adversarial attacks, protecting sensitive data through privacy-preserving techniques and perceptual encryption. My work has been published in reputed venues, including **PETS, ICDAR, and IEEE Transactions**. I have also received competitive research fellowships from TIH and NFOBC. I am seeking a **postdoctoral research position** to contribute to the development of secure, robust, and trustworthy AI systems, particularly in foundation models and large language model security.

EDUCATION

Ph.D. in Computer Science (Ongoing) *2022 – Present*
Department of Computer Science and Engineering
Indian Institute of Technology (IIT) Bhilai, India
Cumulative CGPA (till date): 9.48/10
Thesis Focus: Adversarial Security Evaluation & Privacy-Preserving Deep Learning

M.Sc. in Computer Science and Application *2018 – 2020*
Atal Bihari Vajpayee Vishwavidyalaya, Bilaspur, India
Percentage: 79.85%

B.Sc. in Computer Science and Application *2015 – 2018*
Bilaspur University, Bilaspur, India
Percentage: 68.86%

FELLOWSHIPS & AWARDS

- **TIH PhD Fellowship:** Awarded by the Technology Innovation Hub for research on AI-driven GST fraud detection (Dec 2023 – May 2025).
- **UGC-NET:** Qualified six times (2019–2025).
- **NFOBC Doctoral Fellowship:** Awarded by the University Grants Commission (2023, 2024).
- **CG-SET:** Qualified the Chhattisgarh State Eligibility Test (2019).
- **Conference Travel Grants:** Received grants from IndoML Symposium (2024, 2025) and CVIP 2025 to attend conferences and present research.
- **ANRF International Travel Grant:** Awarded to attend ICDAR 2025 and present research.

RESEARCH EXPERIENCE

Project Scientist *Jan 2023 – Dec 2023*
Indian Institute of Technology (IIT) Delhi

- Evaluated next-generation wireless systems using ML-driven optimization.
- Validated performance metrics and prepared grant-compliance documentation.

TEACHING EXPERIENCE

Teaching Assistant *2023 – Present*
Indian Institute of Technology (IIT) Bhilai

- Conducted tutorials, designed problem sets and coding assignments, addressed student queries, and evaluated assignments and examinations.
- **Courses:**
  (a) Introduction to Programming (CSL-100), 2023-M, 2024-M.
  (b) Adversarial Machine Learning (CSL-607), 2024-W, 2025-W, 2026-W.
  (c) AI/ML Lab (DSP-301), 2026-M.

Guest Faculty (Computer Science) *2020 – 2022*
Atal Bihari Vajpayee University

- Taught Machine Learning, Data Structures, and Programming.

- Mentored undergraduate capstone projects in applied machine learning.

PUBLICATIONS

Published

1. S. K. Padhi, A. Tiwari, **U. Kashyap**, S. S. Ali, *"Doc-Protector: A Self-Healing Approach for Digital Documents."* **The 20th International Conference on Document Analysis and Recognition (ICDAR)**, 2026.
2. **U. Kashyap**, S. S. Ali, *"GAN-Invert: Unveiling Vulnerabilities in Privacy-Preserving Facial Transformations."* **Proceedings on Privacy Enhancing Technologies (PETs)**, 2026.
3. **U. Kashyap**, A. Vishwakarma, S. S. Ali, *"Generalize Adversarial Malware Detection."* **Journal of Hardware and Systems Security**, 10(1), p.11, 2026.
4. R. Kumar, **U. Kashyap**, S. S. Ali, *"Gradient-Guided Adversarial Patch Attack for Deep Neural Networks."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 227–245, 2026.
5. **U. Kashyap**, S. K. Padhi, S. S. Ali, *"Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods."* **IEEE Transactions on Artificial Intelligence**, 2025.
6. A. Vishwakarma, **U. Kashyap**, S. S. Ali, *"Adversarial Malware Detection."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, pp. 277–286, 2024.

Under Publication

7. S. K. Padhi, D. Diwedi **U. Kashyap**, S. S. Ali, *"MIRROR: Makeup Identity Restoration for RObust Facial Recognition."* **International Conference on Computer Vision and Image Processing (CVIP)**, 2026. **Accepted**
8. **U. Kashyap**, R. Borikar, S. K. Padhi, S. S. Ali, *"The Cosine Conspiracy: Subverting CLIPure-Cos with a Latent-Space Informed Attack."* **International Conference on Security, Privacy, and Applied Cryptography Engineering (SPACE)**, 2026. **Accepted**
9. S. Nevase, **U. Kashyap**, S. S. Ali, *"Perceptually-Guided Adversarial Diffusion: Enhancing Visual Fidelity in Facial Privacy."* **International Conference on Computer Vision and Image Processing (CVIP)**, 2025. **Under Publication**

Preprints

10. S. K. Padhi, H. Kumar, **U. Kashyap**, S. S. Ali, *"De-Fake: Style-Based Anomaly Deepfake Detection."* preprint arXiv:2507.03334, 2025.
11. S. M. Shahid, S. K. Padhi, **U. Kashyap**, S. S. Ali, *"Generalized Deepfake Attribution."* preprint arXiv:2406.18278, 2024.

CONFERENCE PRESENTATIONS

Oral Presentations

- **PETS 2026 (Online):** GAN-Invert: Unveiling Vulnerabilities in Privacy-Preserving Facial Transformations
- **ICDAR 2025:** Doc-Protector: A Self-Healing Approach for Digital Documents
- **CVIP 2025:** Perceptually-Guided Adversarial Diffusion: Enhancing Visual Fidelity in Facial Privacy
- **SPACE 2024:** Adversarial Malware Detection

Poster Presentations

- **IndoML 2025:** Is Perceptual Encryption Secure? A Security Benchmark for Perceptual Encryption Methods

TECHNICAL SKILLS

Languages	Python, C, C++, LaTeX
Tools & Platforms	Git, Docker, Linux, Google Colab, Jupyter, Hugging Face
Core Competencies	Model Robustness, Adversarial Attack/Defense, Computer Vision

LANGUAGES

Hindi: Native **English:** Professional